Listing 17.8 A step-scaling policy adding one more instance to an Auto Scaling group

```
# [...]
ScalingUpPolicy:
  Type: 'AWS::AutoScaling::ScalingPolicy'                ①
  Properties:
    AdjustmentType: 'ChangeInCapacity'                   ②
    AutoScalingGroupName: !Ref AutoScalingGroup          ③
    PolicyType: 'StepScaling'                            ④
    MetricAggregationType: 'Average'                     ⑤
    EstimatedInstanceWarmup: 60                          ⑥
    StepAdjustments:                                     ⑦
    - MetricIntervalLowerBound: 0                        ⑧
      ScalingAdjustment: 1                               ⑨
# [...]
```

① Creates a scaling policy

② The scaling policy increases the capacity by an absolute number.

③ Attaches the scaling policy to the Auto Scaling group

④ Creates a scaling policy of type step scaling

⑤ The aggregation type used when evaluating the steps, based on the metric defined within the CloudWatch alarm that triggers the scaling policy

⑥ The metrics of a newly launched instance are ignored for 60 seconds while it boots up.

⑦ Defines the scaling steps. We use a single step in this example.

⑧ The scaling step is valid from the alarms threshold to infinity.

⑨ Increases the desired capacity of the Auto Scaling group by 1

To scale down the number of instances when the queue is empty, a CloudWatch alarm and scaling policy with the opposite values needs to be defined.

You're now ready to scale. We've prepared a load test that will quickly generate 250 messages for the URL2PNG application. A virtual machine will be launched to process jobs from the SQS queue. After a few minutes, when the load test is finished, the additional virtual machine will disappear. Update the CloudFormation stack with the following command to start the load test:

```
$ aws cloudformation update-stack --stack-name url2png \
➥ --template-url https:/ /s3.amazonaws.com/\
➥ awsinaction-code3/chapter17/url2png-loadtest.yaml \
➥ --capabilities CAPABILITY_IAM
```

Watch for the following things to happen, with the help of the AWS Management Console: